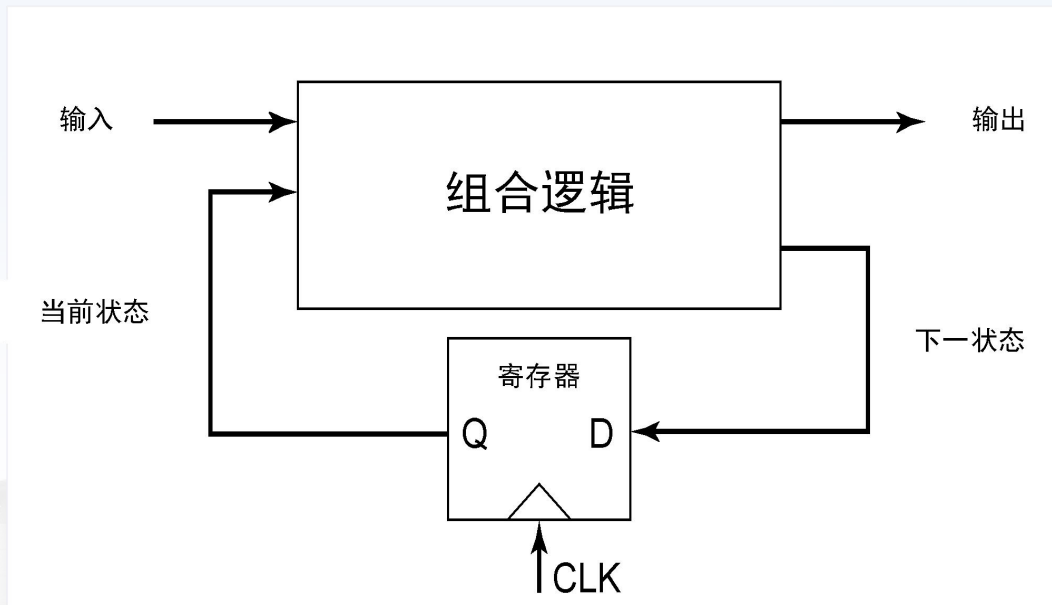




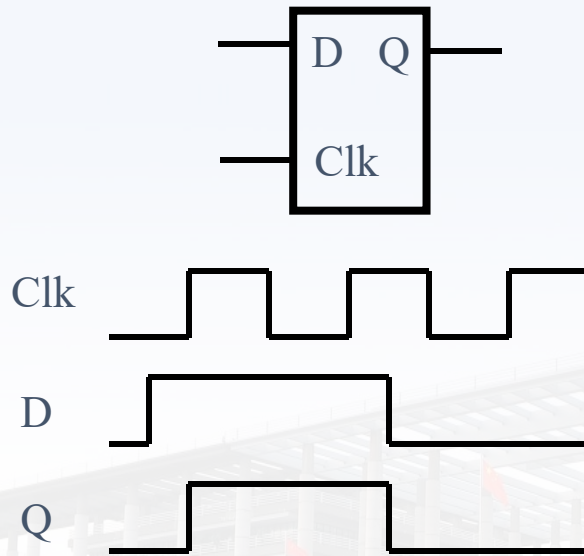
数字集成电路设计--第七章 时序逻辑电路设计



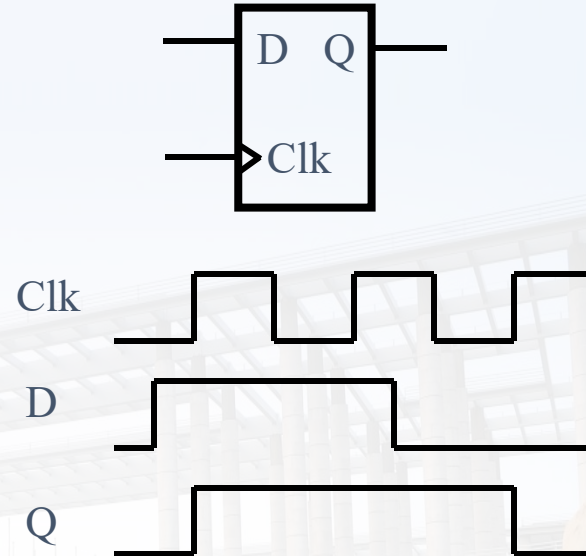
通用有限状态机 (FSM)

- 反馈机制

锁存器和寄存器



- 锁存器
时钟电平触发

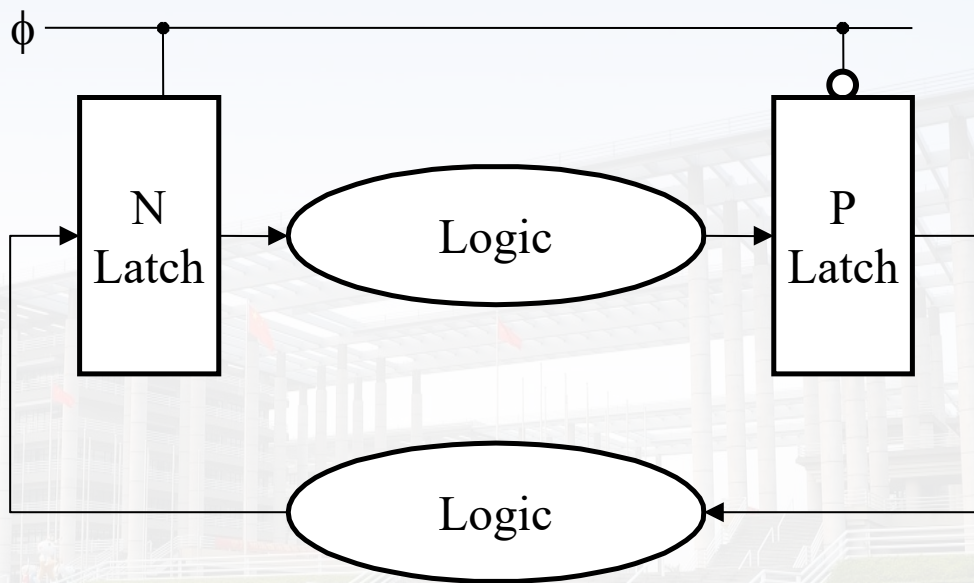


- 寄存器
时钟边沿触发

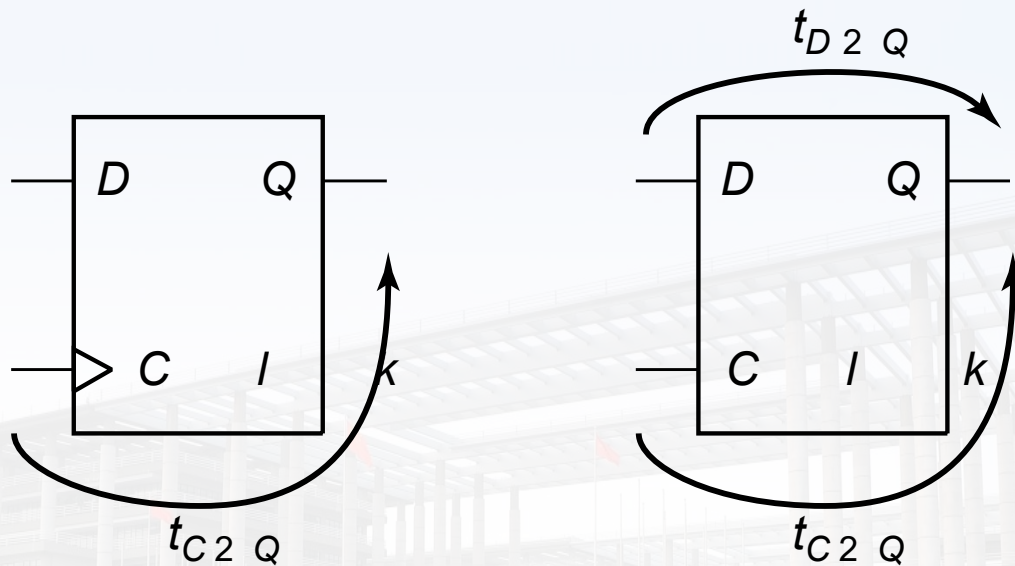
基于锁存器的设计

• $\phi = 1$ 时 N型锁存器导通

• $\phi = 0$ 时 P型锁存器导通



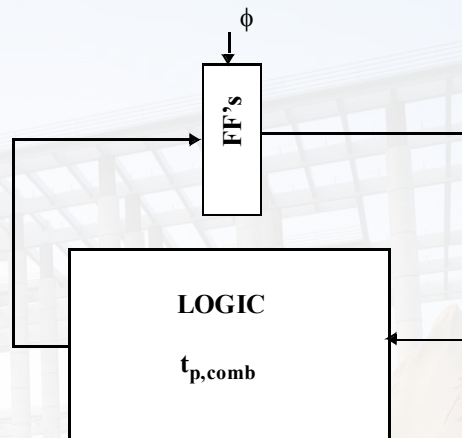
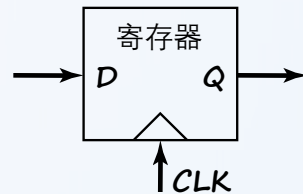
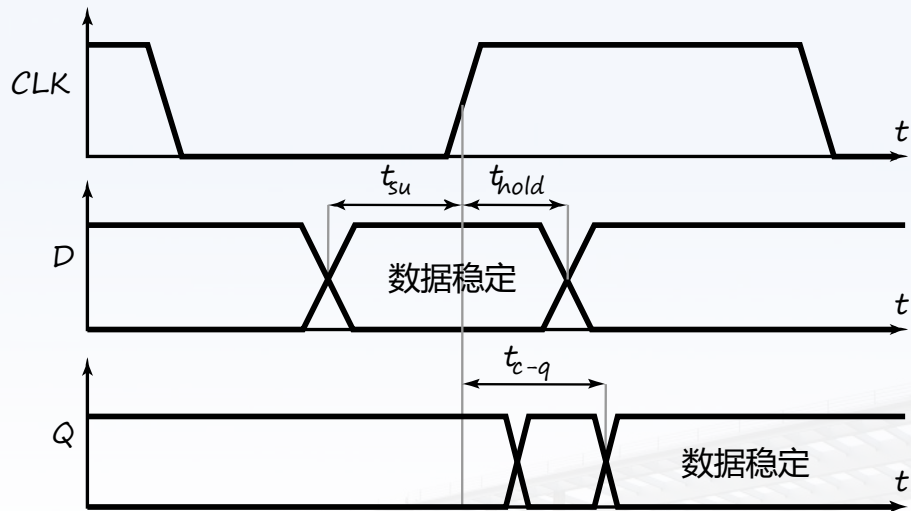
时间特性



寄存器

锁存器

时序分析



时序约束:

$$T \geq t_{c-q} + t_{plogic} + t_{su}$$

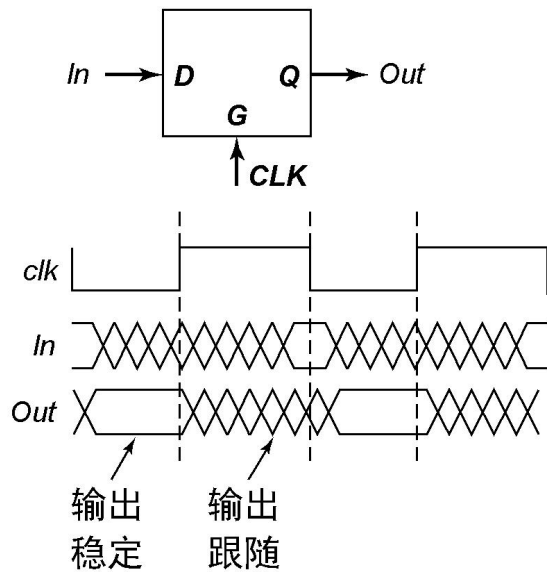
$$t_{cdregister} + t_{cdlogic} \geq t_{hold}$$

存储单元分类

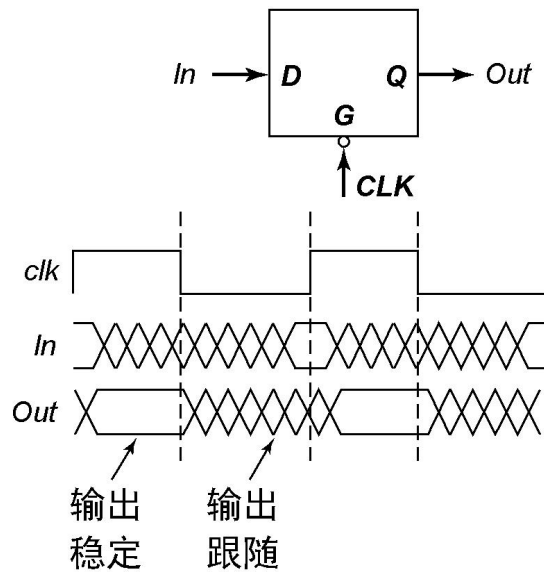
- 层次划分：
 - 前台存储器：嵌入在逻辑中的存储器
 - 后台存储器：集中存储内核
- 存储器特性
 - 静态存储器：数据一致保持存储状态
 - 动态存储器：需要周期刷新，弥补电荷泄露
- 触发模式
 - 锁存器
 - 寄存器

锁存器

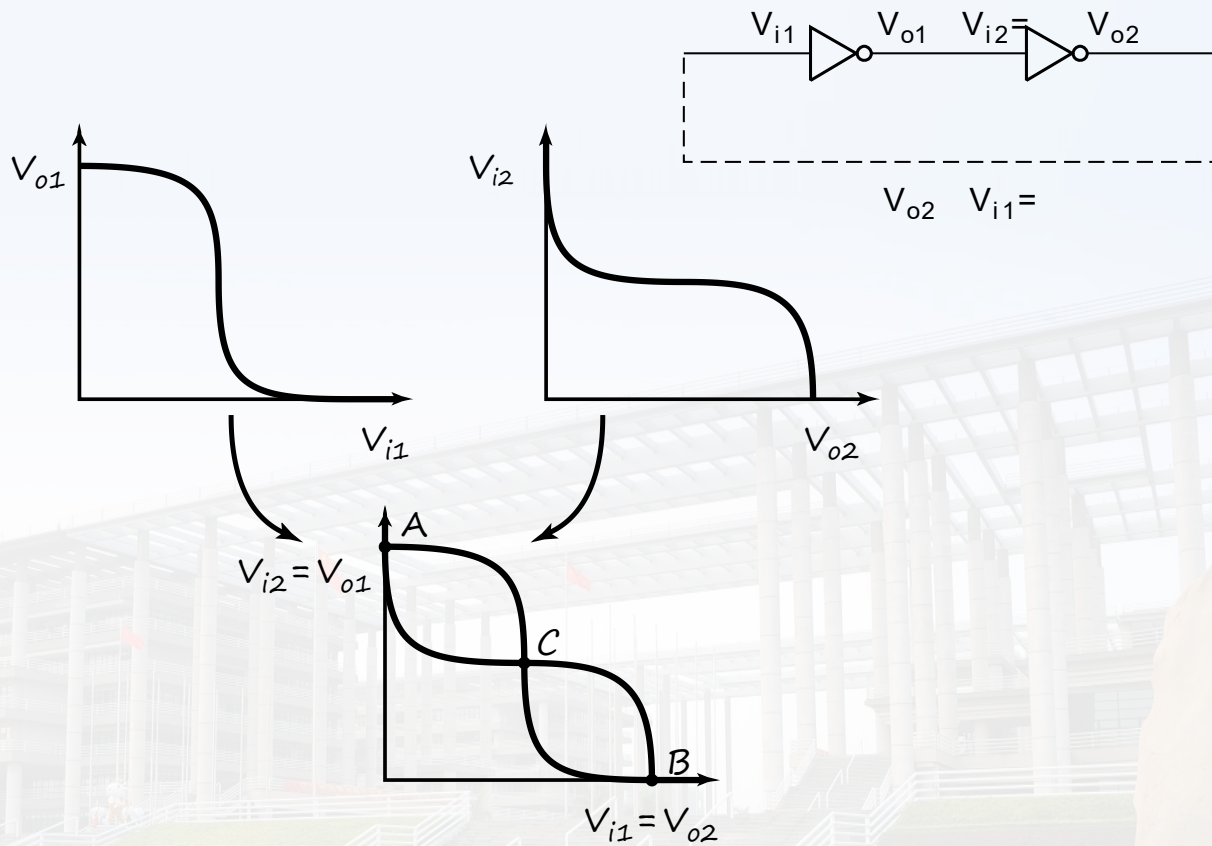
正电平触发



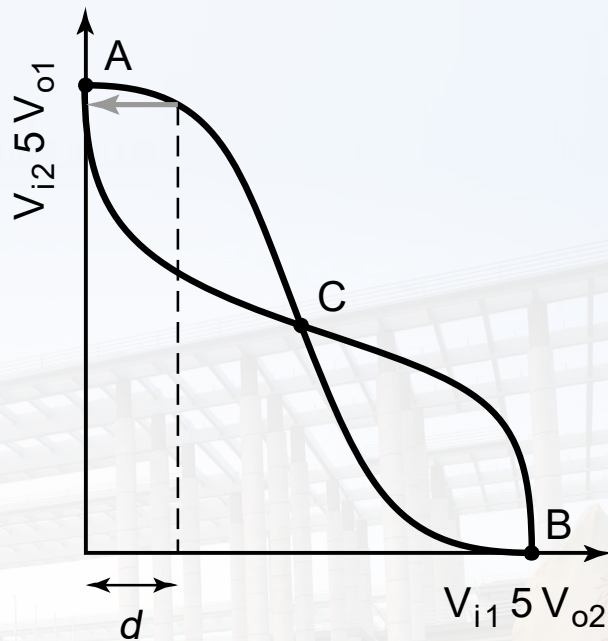
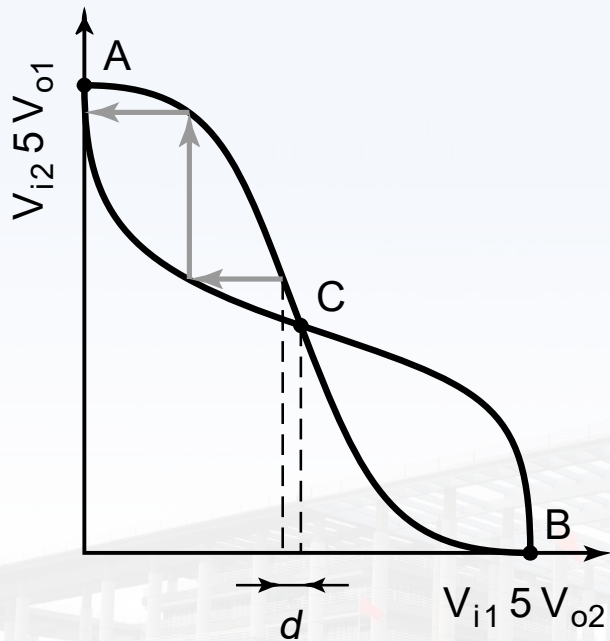
负电平触发



正反馈双稳态



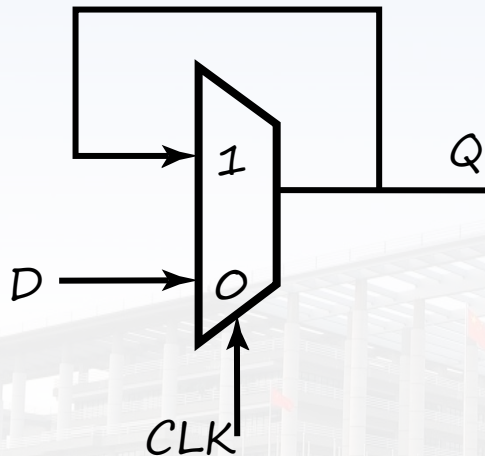
亚稳态电路



增益需要大于1

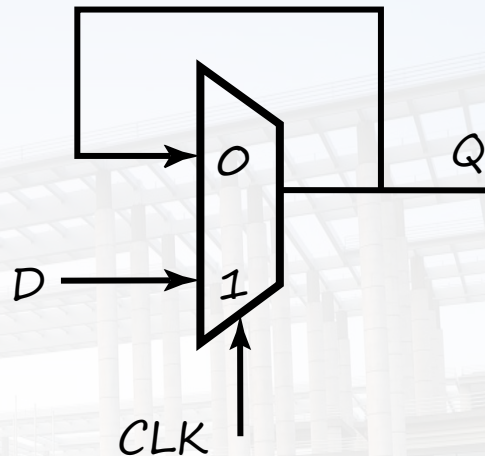
多路开关型锁存器

负锁存器
(CLK = 0)



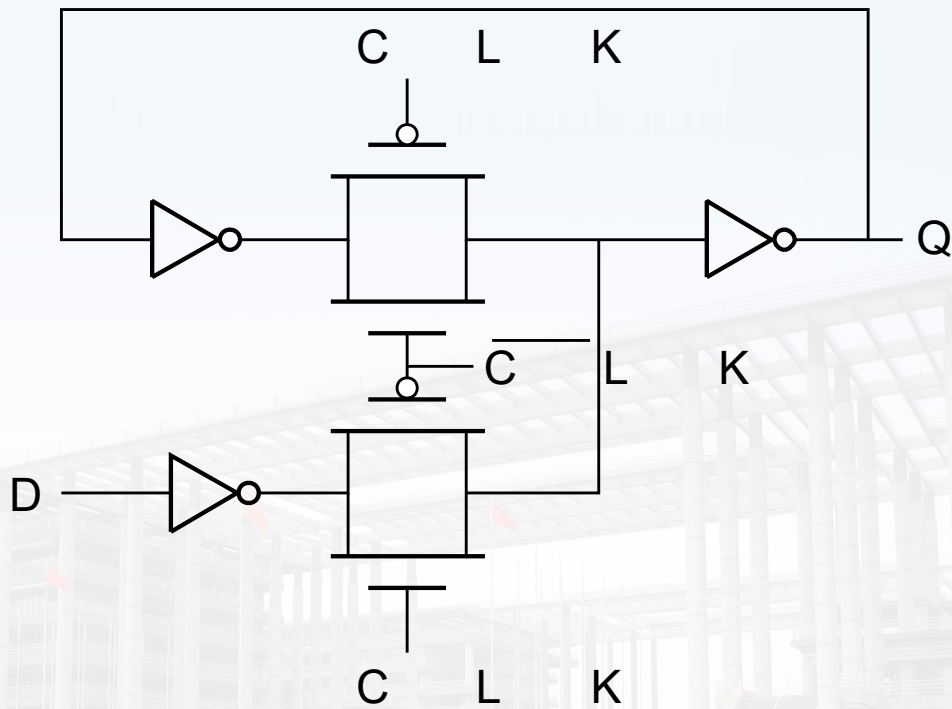
$$Q = \overline{Clk} \cdot Q + Clk \cdot In$$

正锁存器
(CLK = 1)



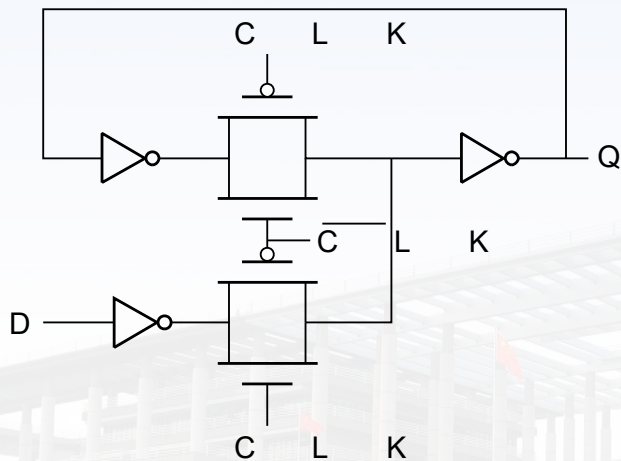
$$Q = Clk \cdot Q + \overline{Clk} \cdot In$$

多路开关型锁存器

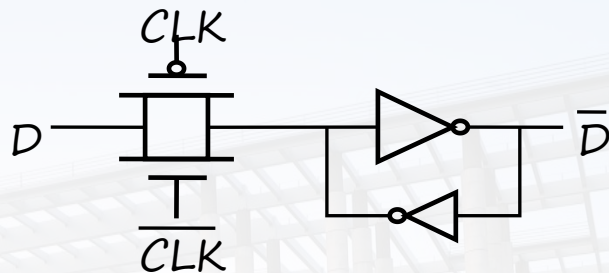


锁存器的写入

时钟信号作为传输门的控制单元

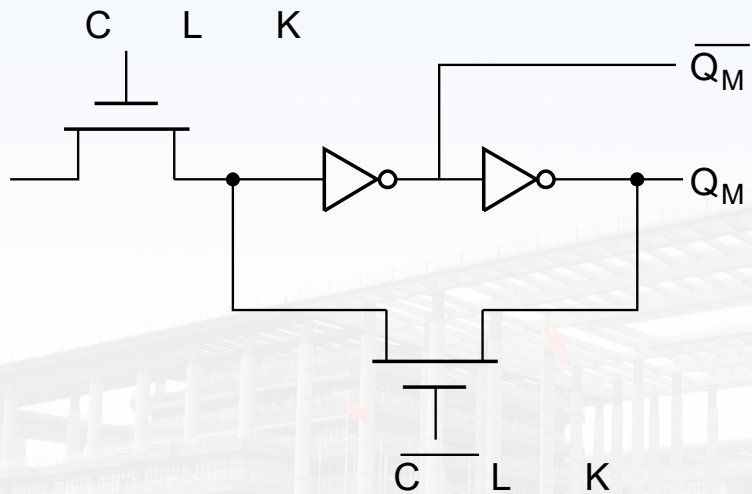


MUX

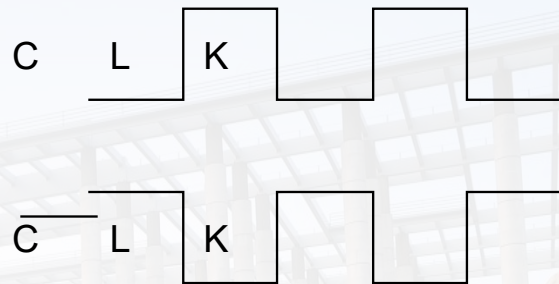


状态保持

多路开关型锁存器

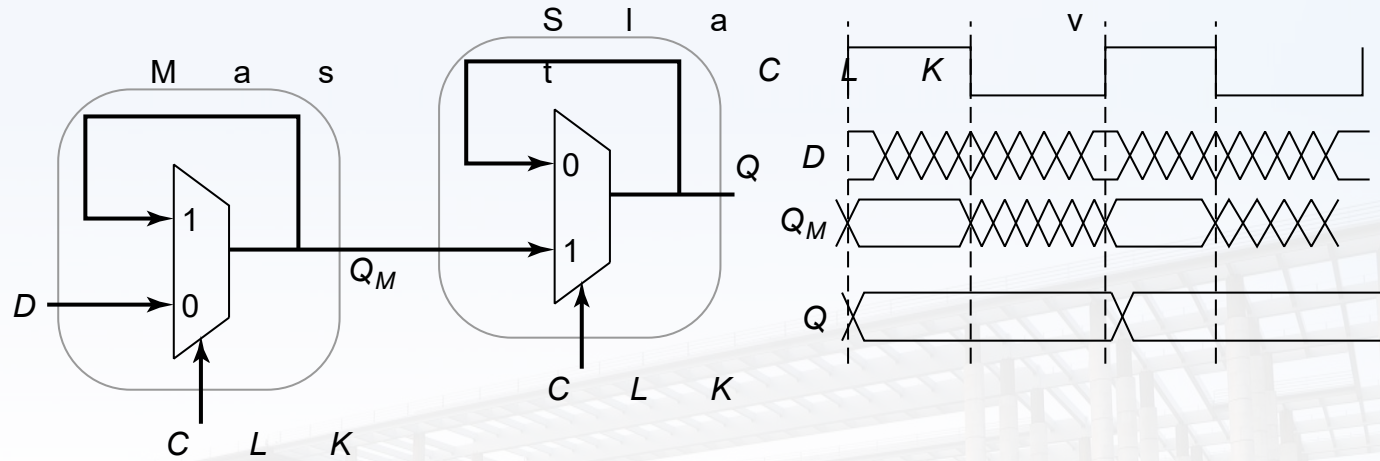


仅NMOS



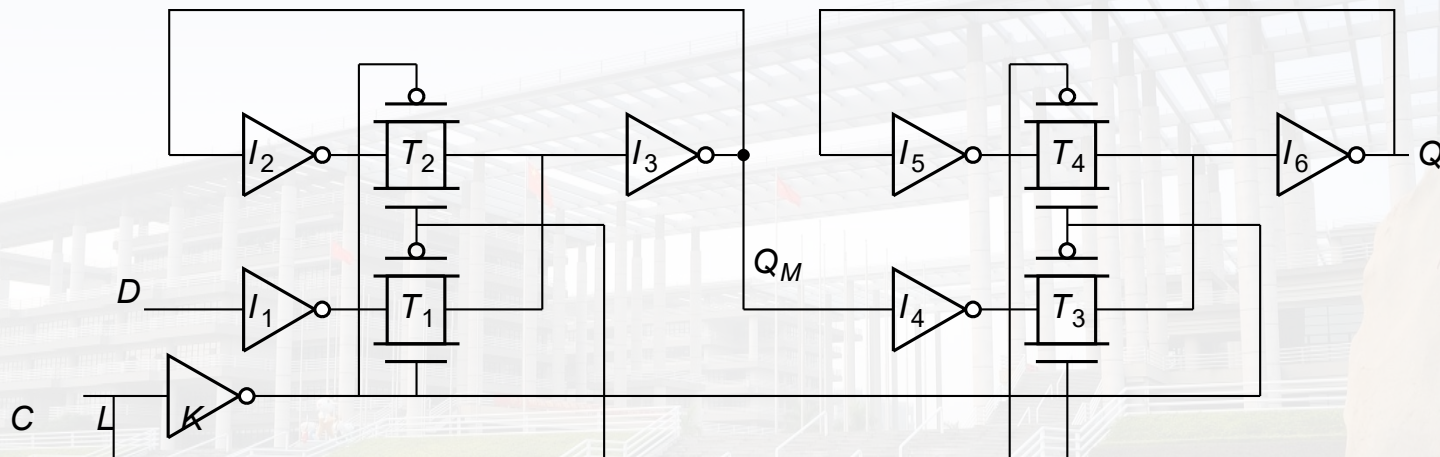
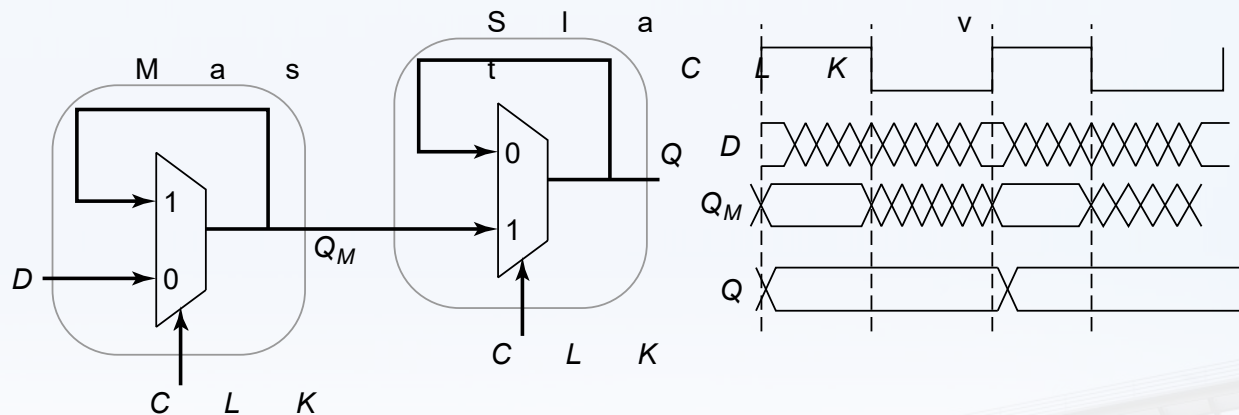
非交叠时钟

主从边沿触发寄存器

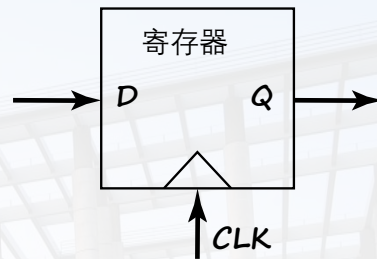
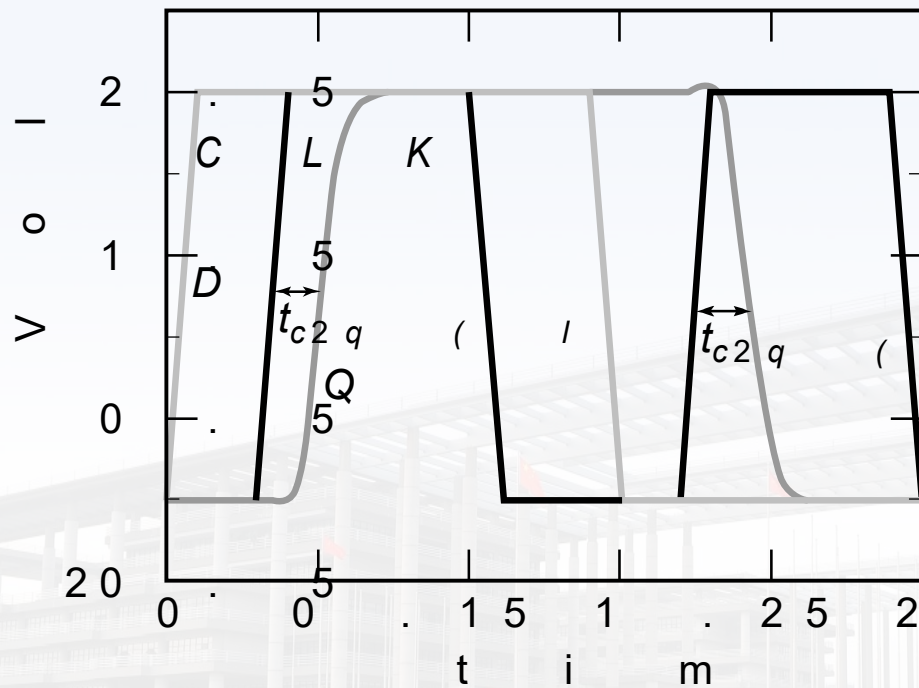


两个反向的锁存器级联构建主从锁存器对

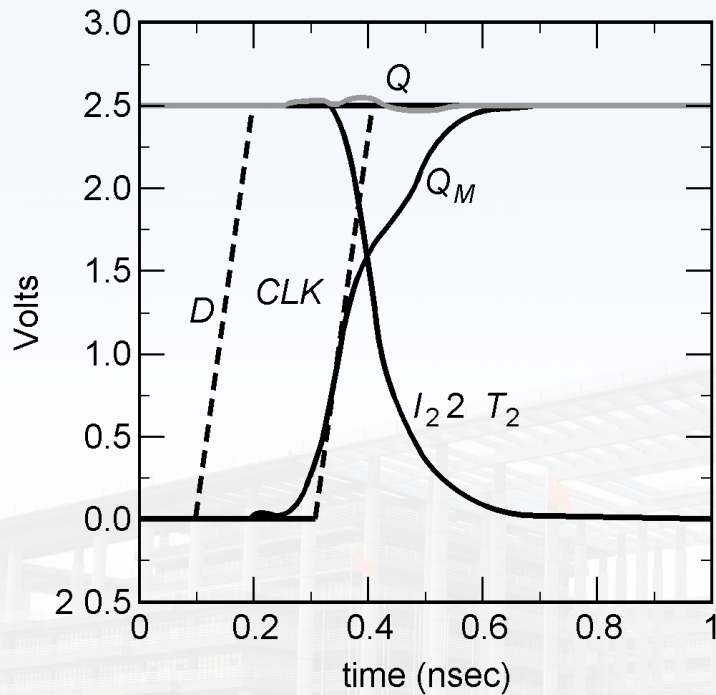
主从正沿触发寄存器



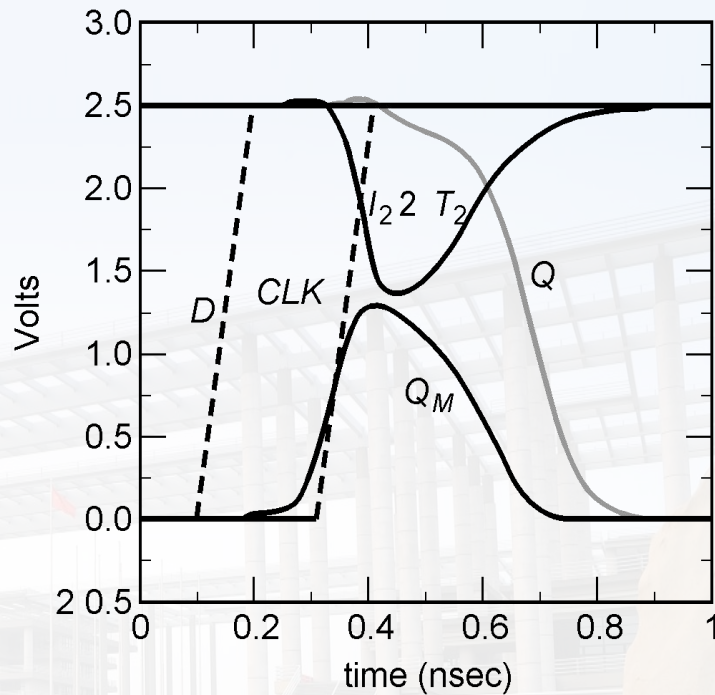
传播延时



建立时间

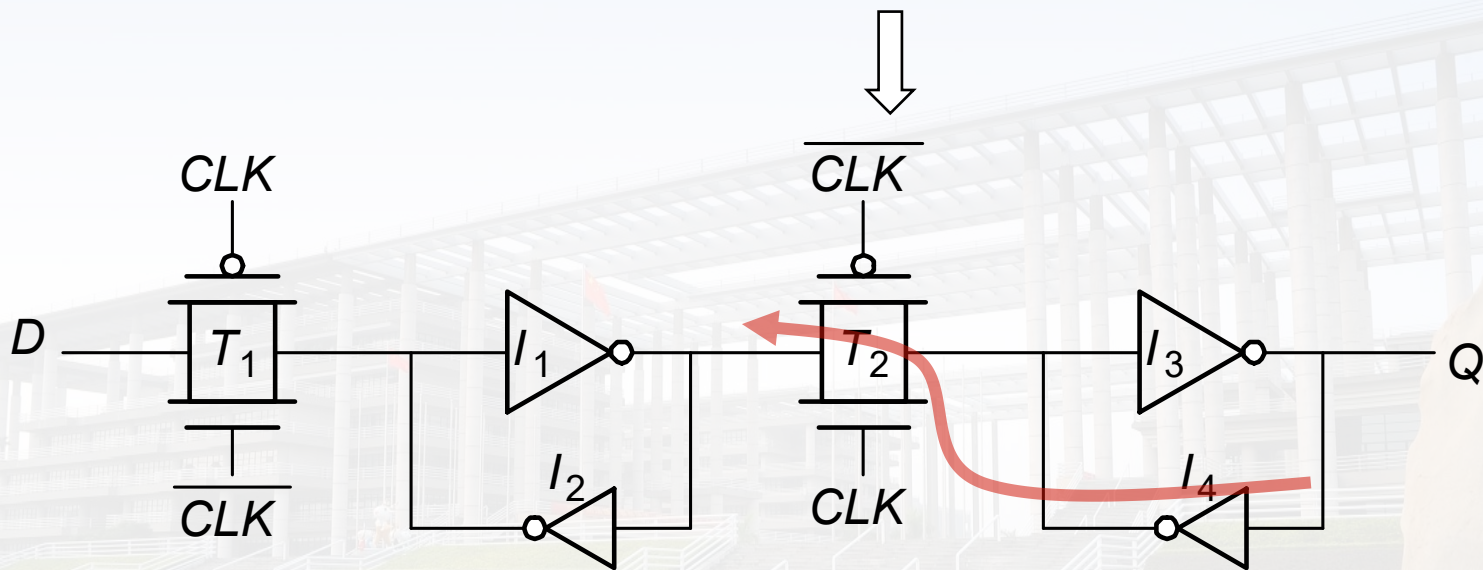
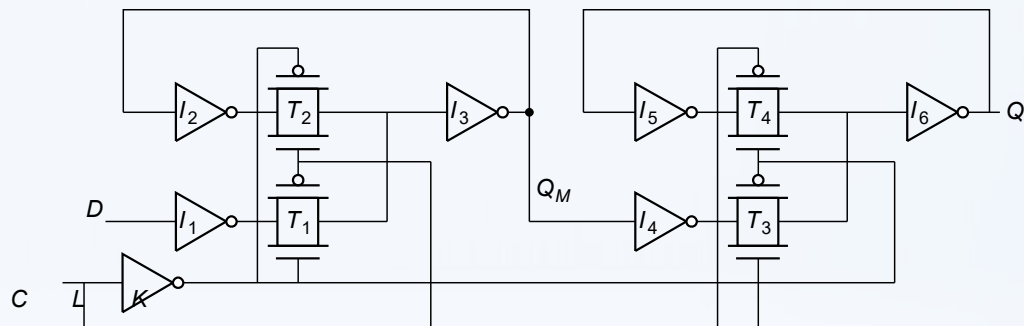


(a) $T_{\text{setup}} = 0.21 \text{ nsec}$

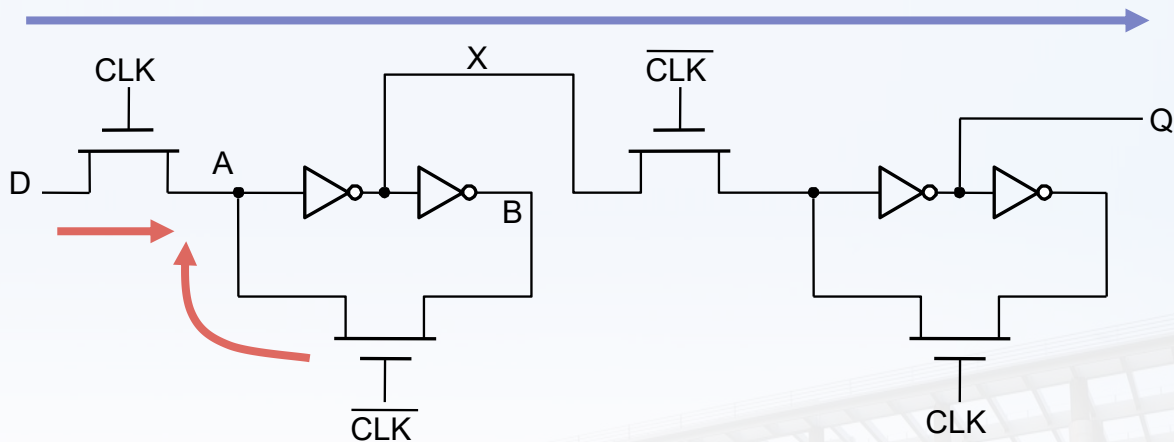


(b) $T_{\text{setup}} = 0.20 \text{ nsec}$

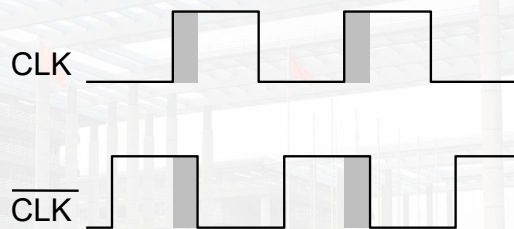
主从寄存器中的电容负载



减少时钟重叠



(a) 电路结构

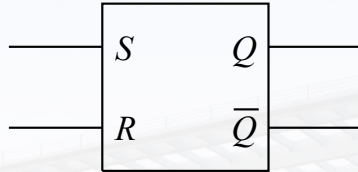
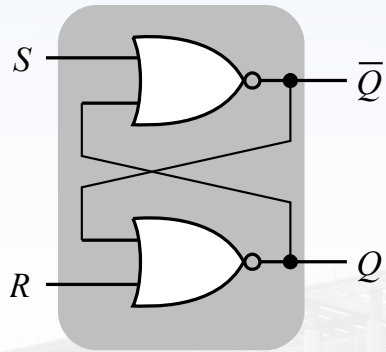


(b) 时钟重叠

1. 竞争问题（上升下降沿均变化）
2. A点的双驱动问题

静态SR触发器

NOR-based SR

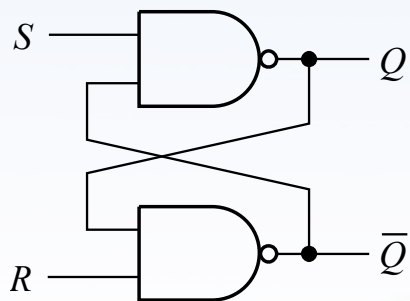


S	R	Q	$\bar{Q}$
0	0	Q	$\bar{Q}$
1	0	1	0
0	1	0	1
1	1	0	0

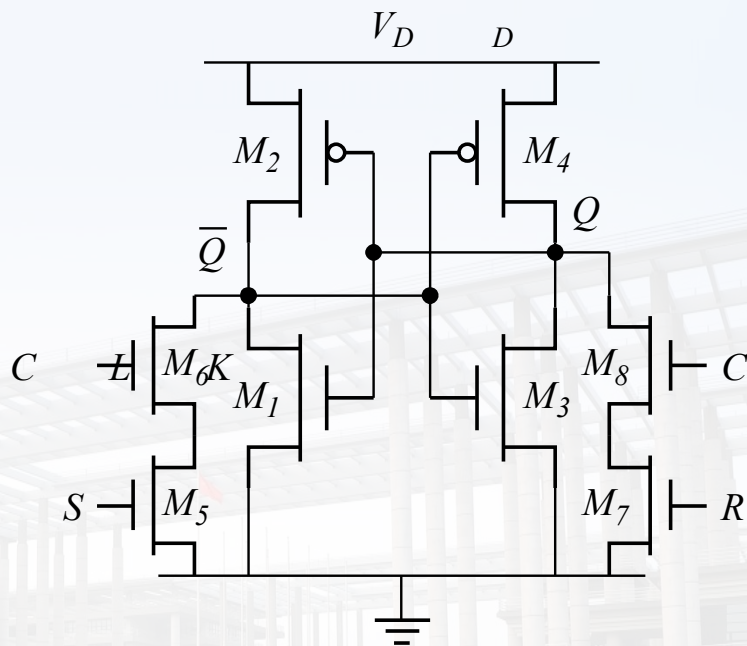
F o r

SR触发器

交叠 NANDs

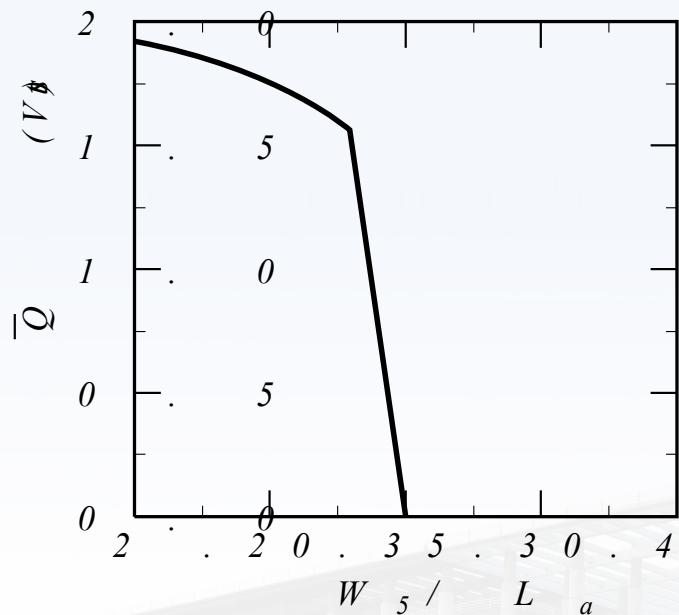


增加时钟信号



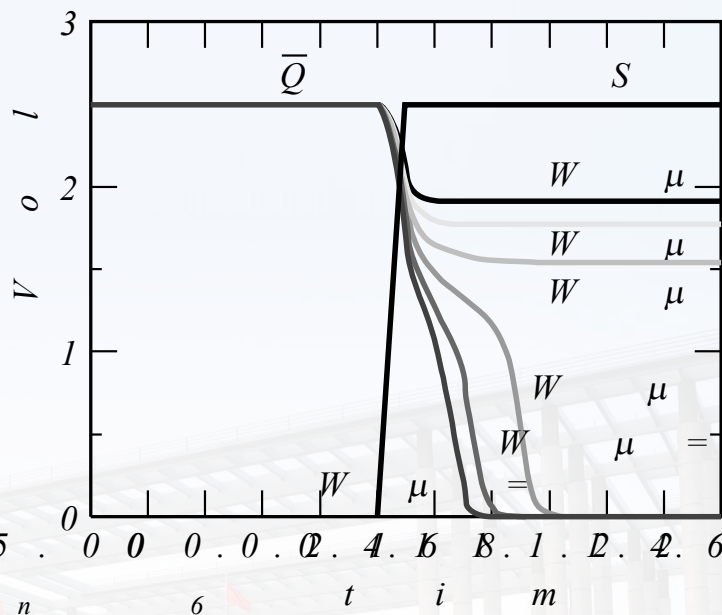
SRAM存储器基本单元

尺寸效应



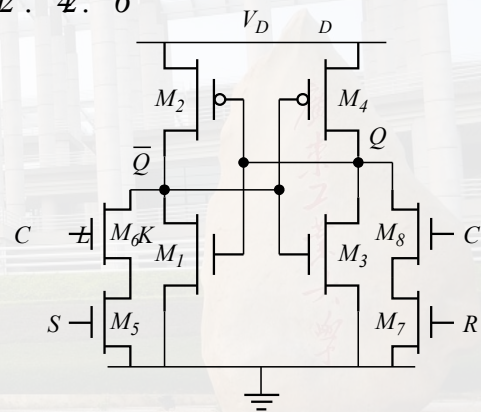
(a)

输出与晶体管尺寸相关



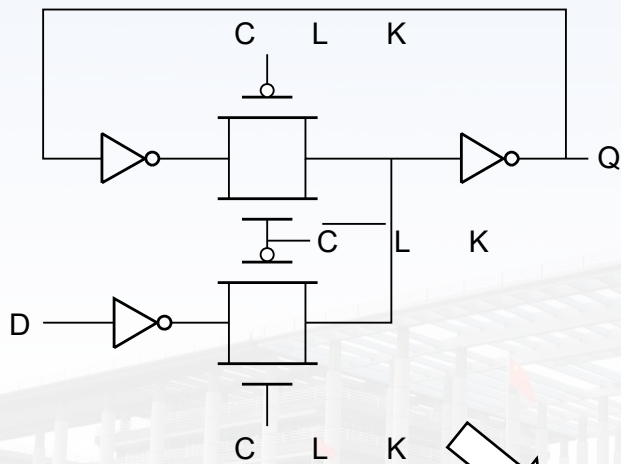
(b)

传输响应

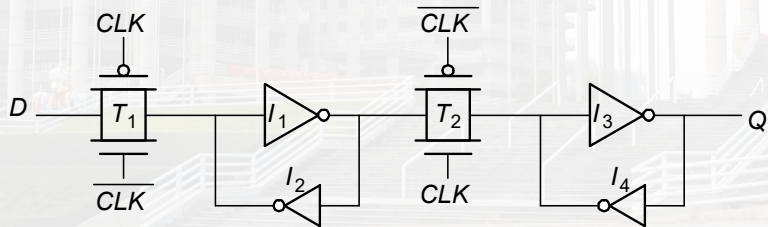
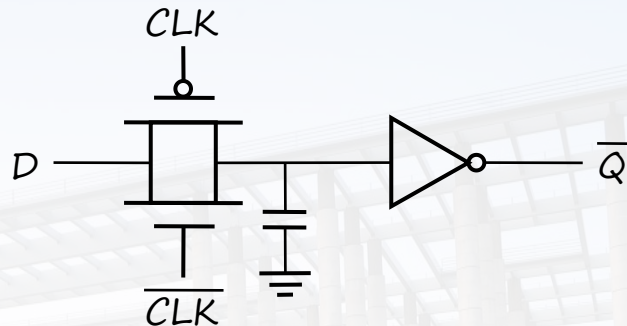


存储机制

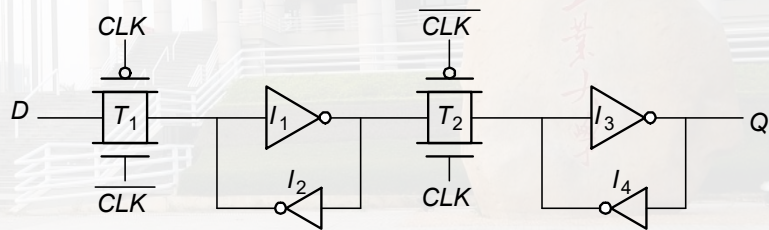
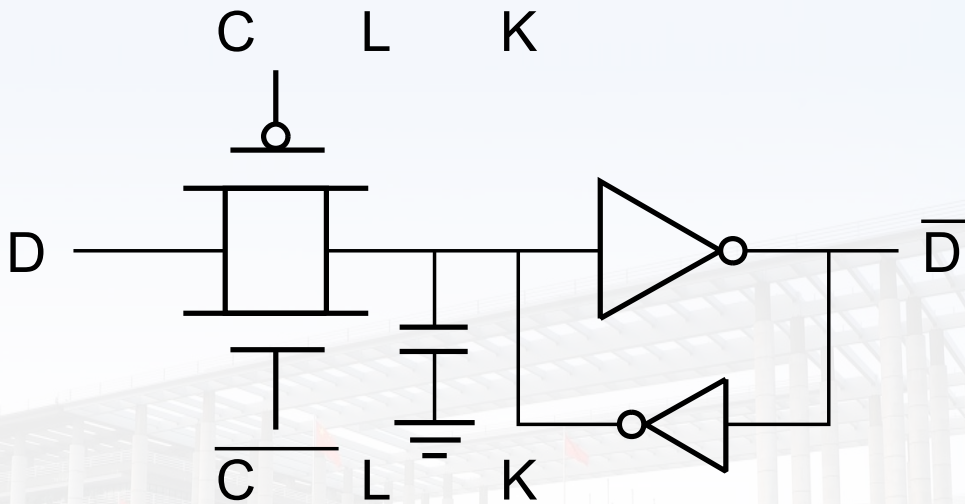
静态



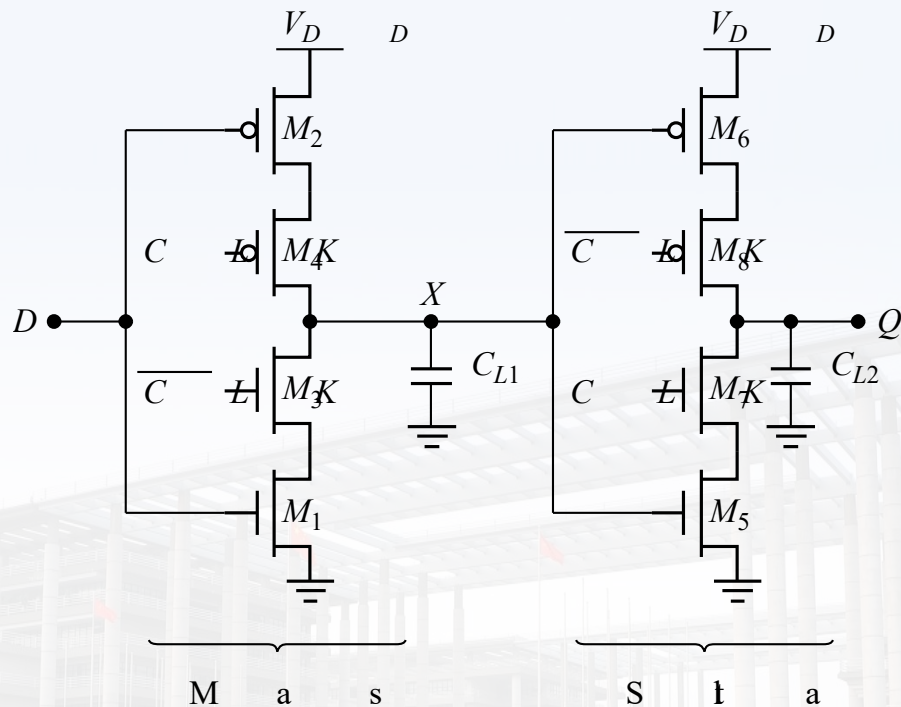
动态



伪静态

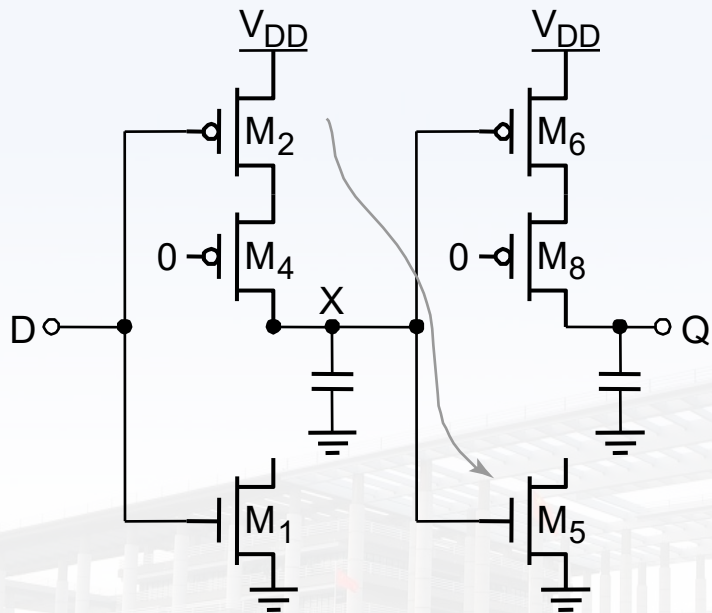


C²MOS (降低对时钟敏感性)

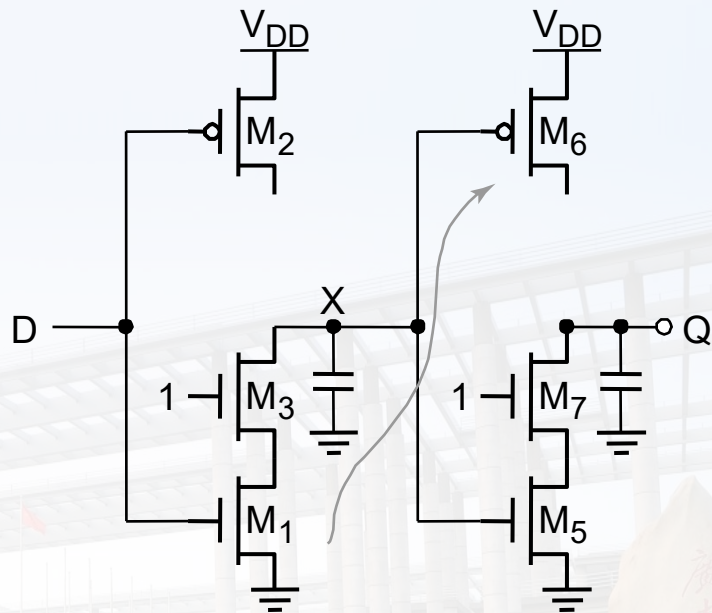


主从正沿触发寄存器

时钟偏差

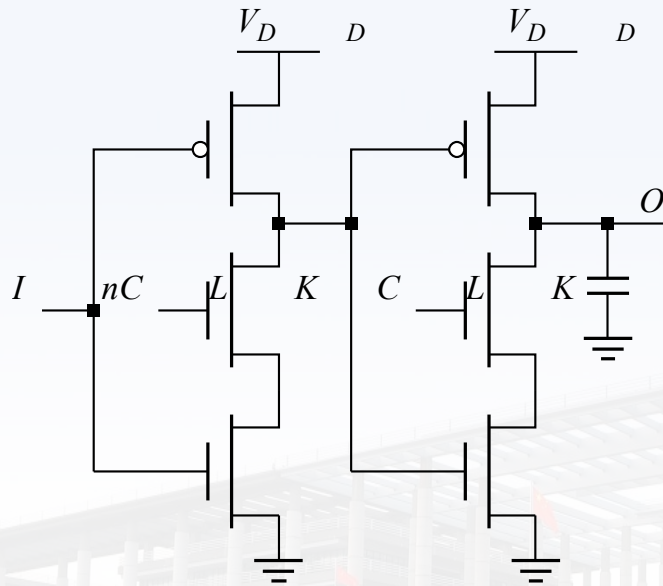


(a) (0-0) 重叠

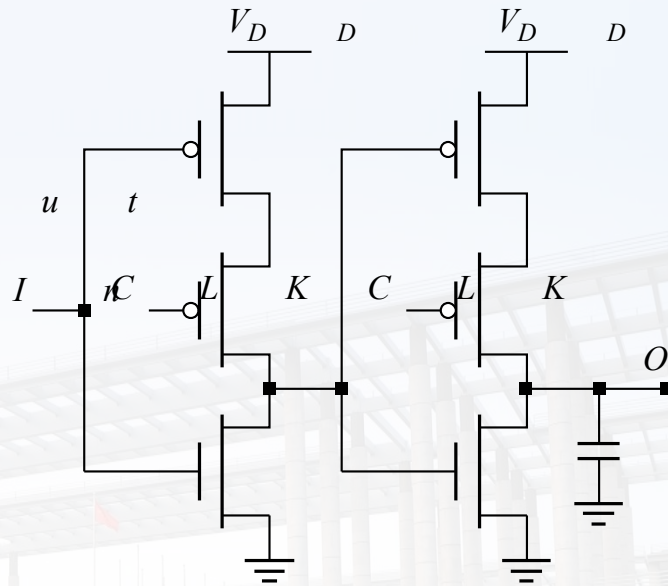


(b) (1-1)重叠

真单相钟控寄存器 (TSPC)

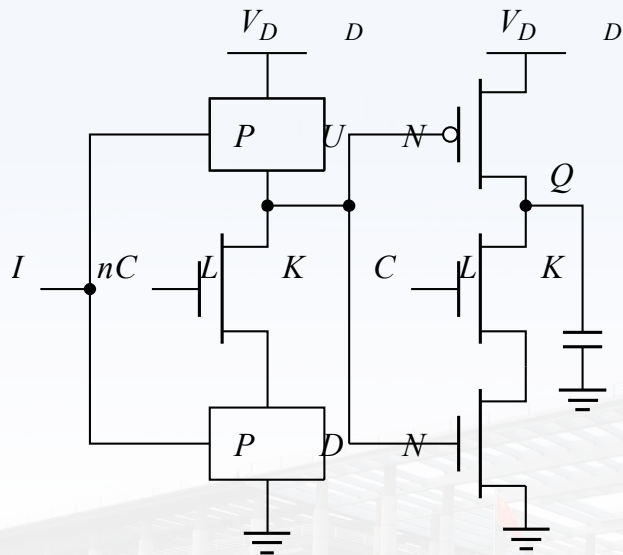


正锁存器
(CLK=1时传递)

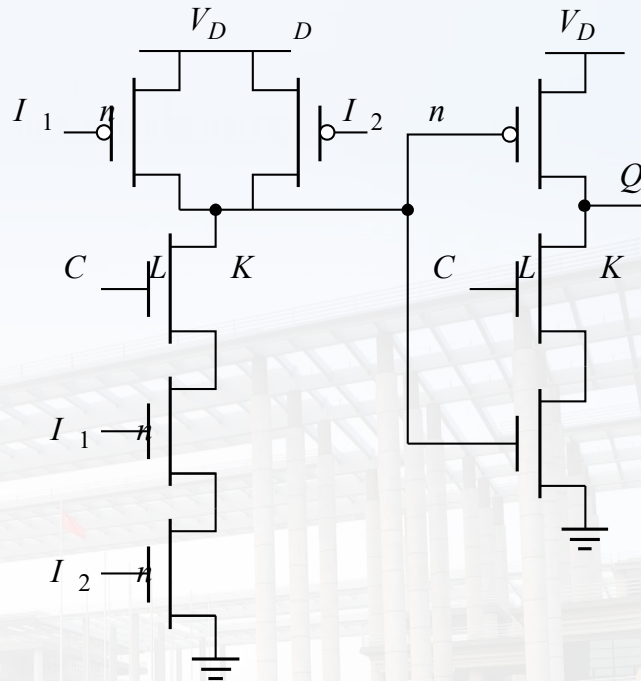


负锁存器
(CLK=0时传递)

TSPC嵌入逻辑

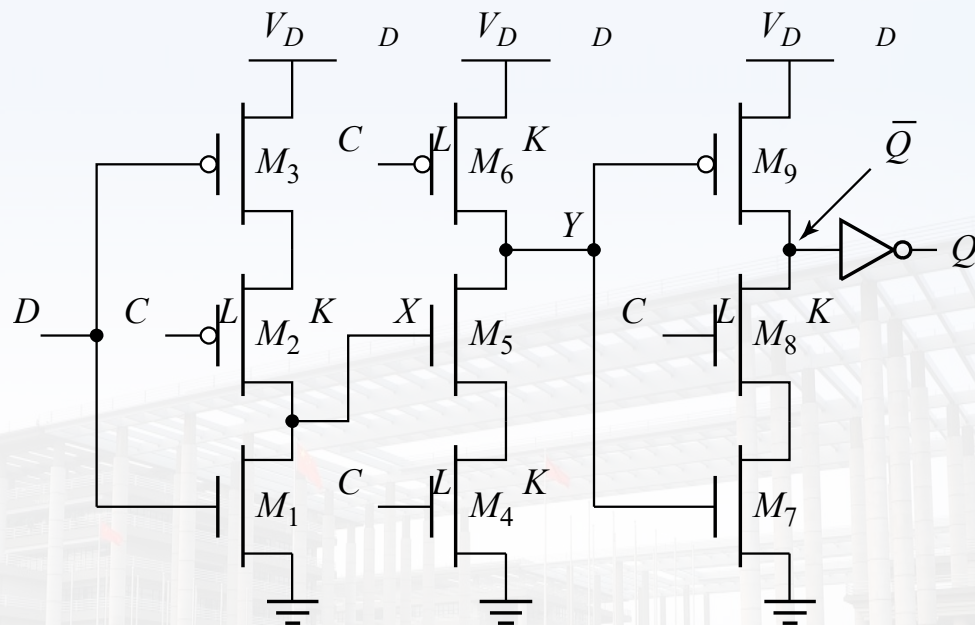


锁存器中包含逻辑

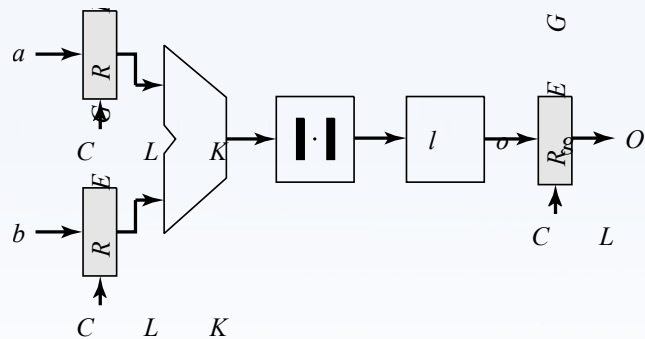


AND 锁存器

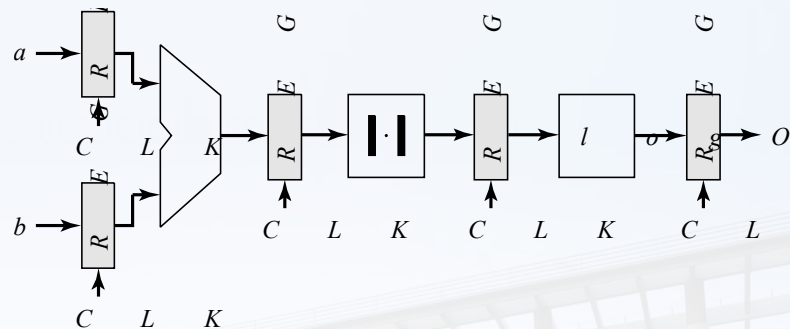
TSPC寄存器



流水线



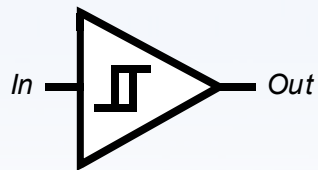
原始



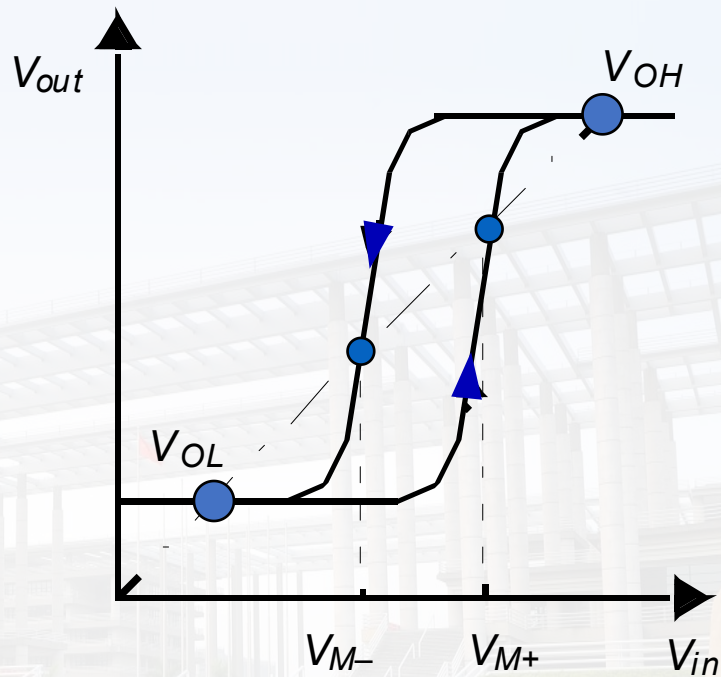
流水化

Clock Period	Adder	Absolute Value	Logarithm
1	$a_1 + b_1$		
2	$a_2 + b_2$	$ a_1 + b_1 $	
3	$a_3 + b_3$	$ a_2 + b_2 $	$\log(a_1 + b_1)$
4	$a_4 + b_4$	$ a_3 + b_3 $	$\log(a_2 + b_2)$
5	$a_5 + b_5$	$ a_4 + b_4 $	$\log(a_3 + b_3)$

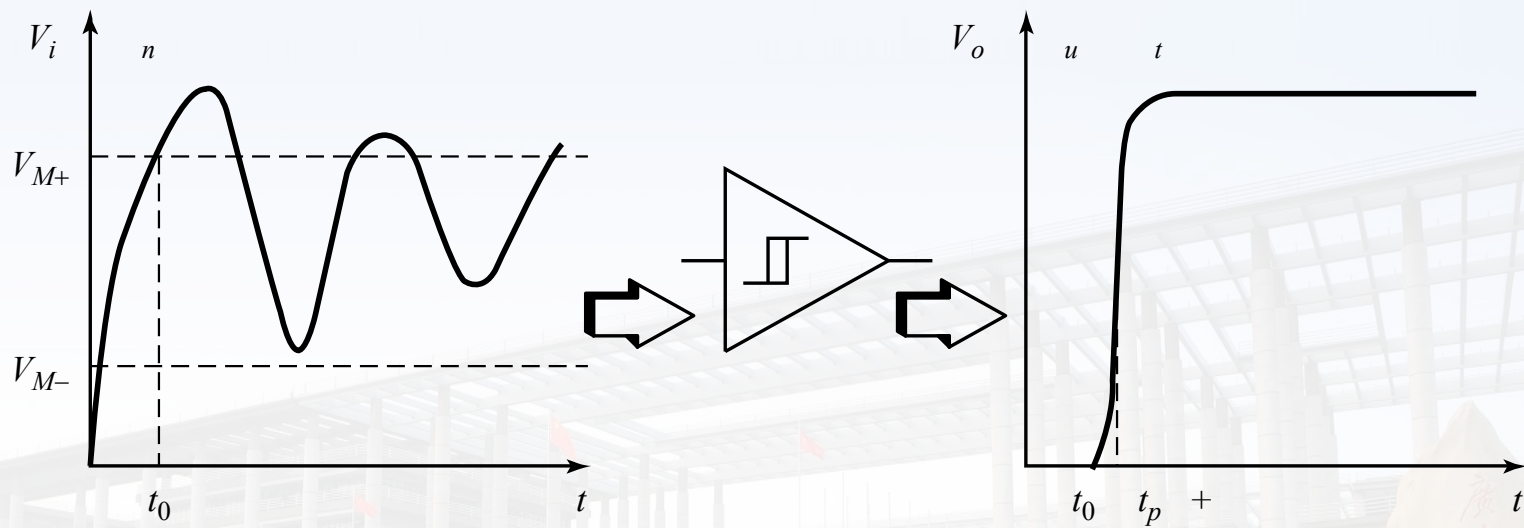
施密特触发器



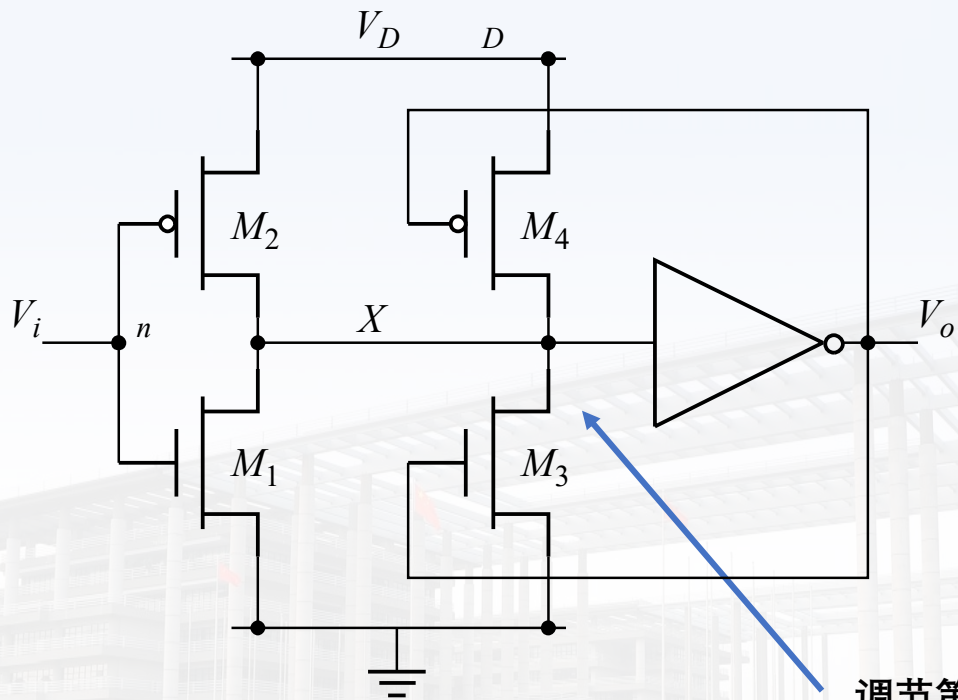
- 输出端快速翻转
- 传输特性有方向性



噪声抑制

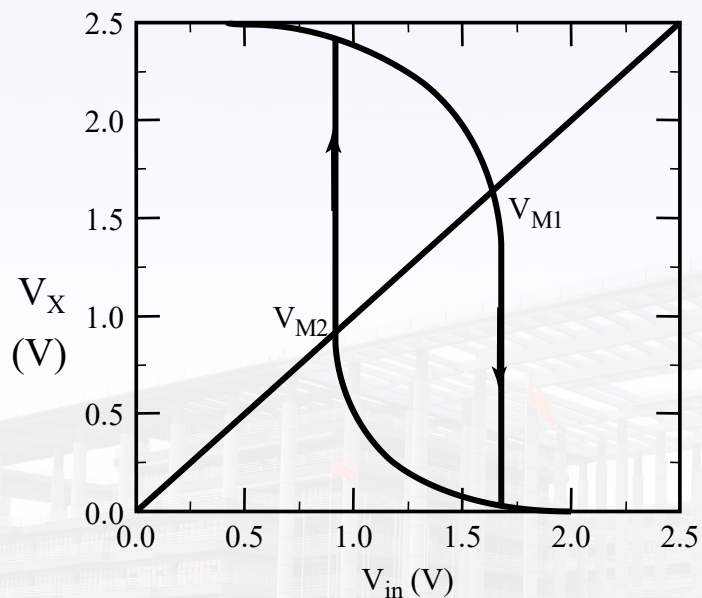
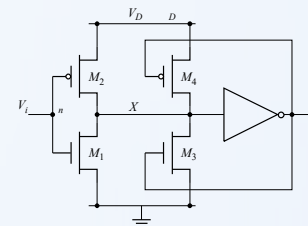


CMOS施密特触发器

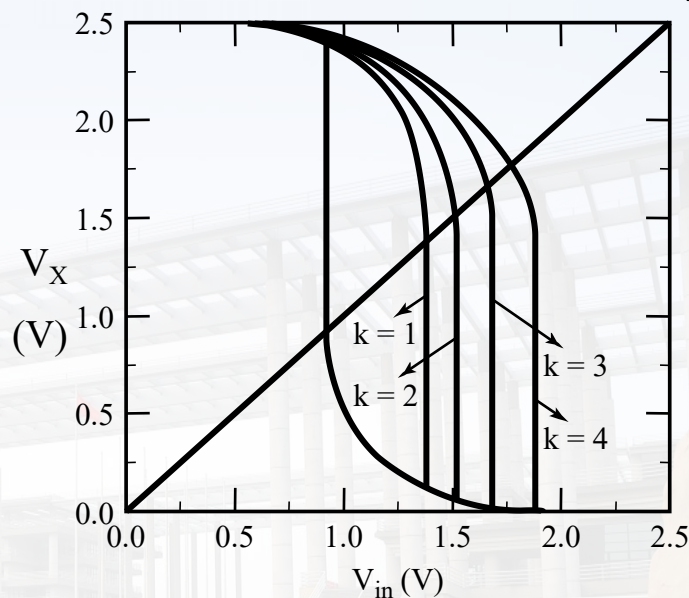


调节第一级方向器的阈值

施密特触发器VTC



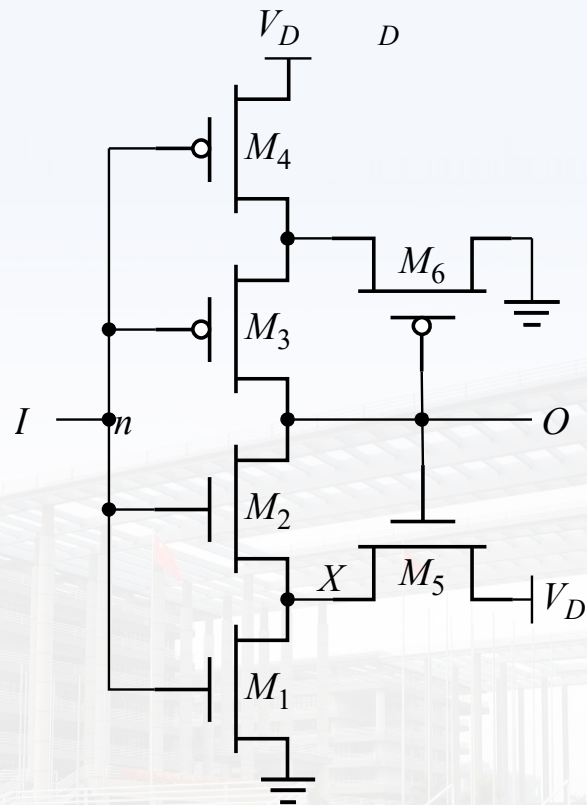
帶有滯環的电压传输特性



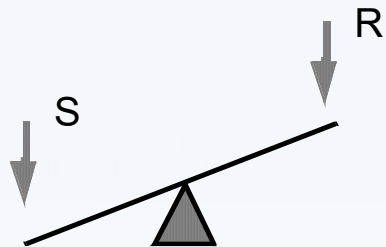
改变PMOS M_4 器件尺寸比的影响

$k \propto 0.5\mu\text{m}$

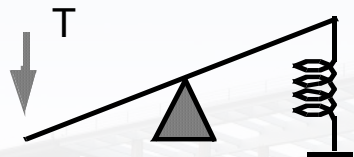
CMOS施密特触发器



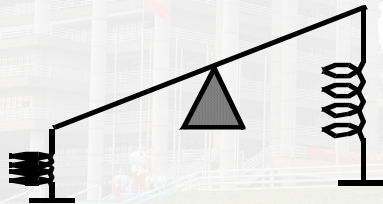
电路结构



双稳态电路
寄存器、施密特触发器

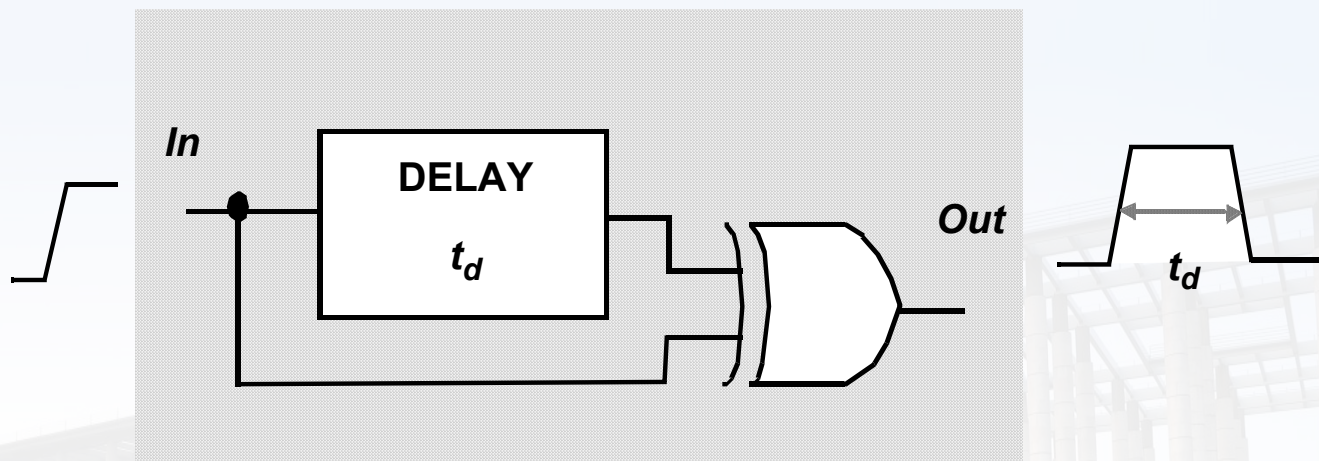


单稳态电路

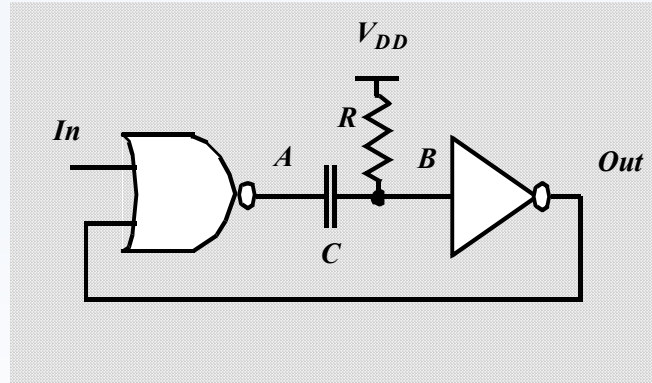


不稳电路
振荡器

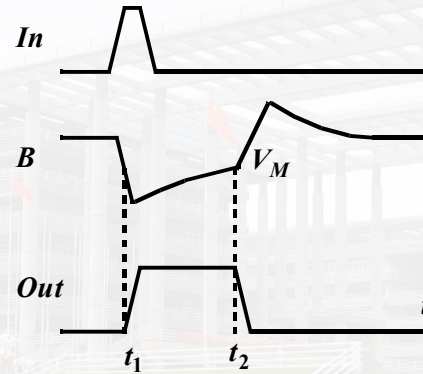
单稳态时序电路



单稳态触发器 (RC-based)

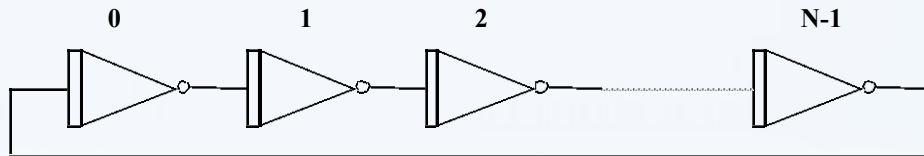


(a) Trigger circuit.

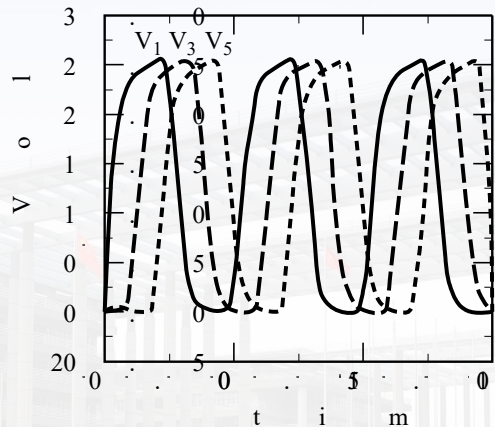


(b) Waveforms.

环形振荡器 (OSC)

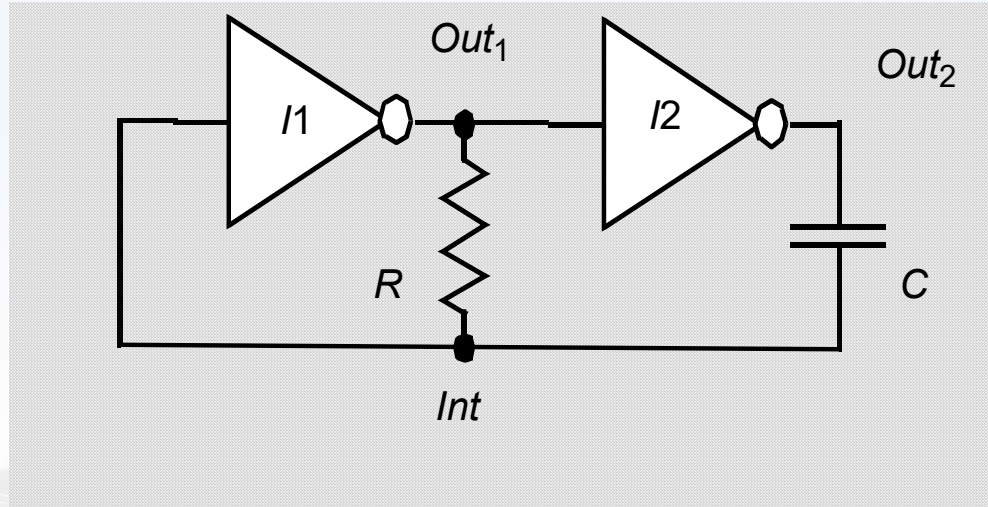


环形振荡器



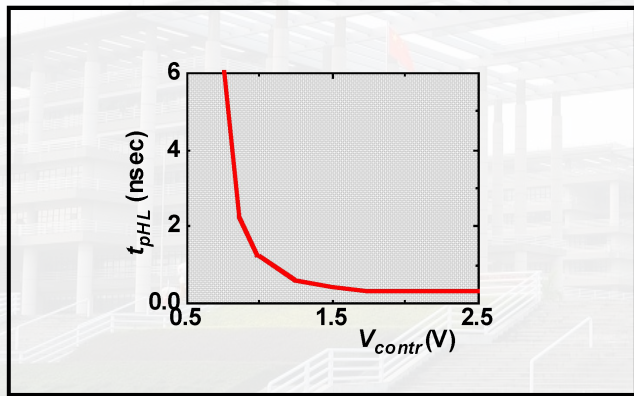
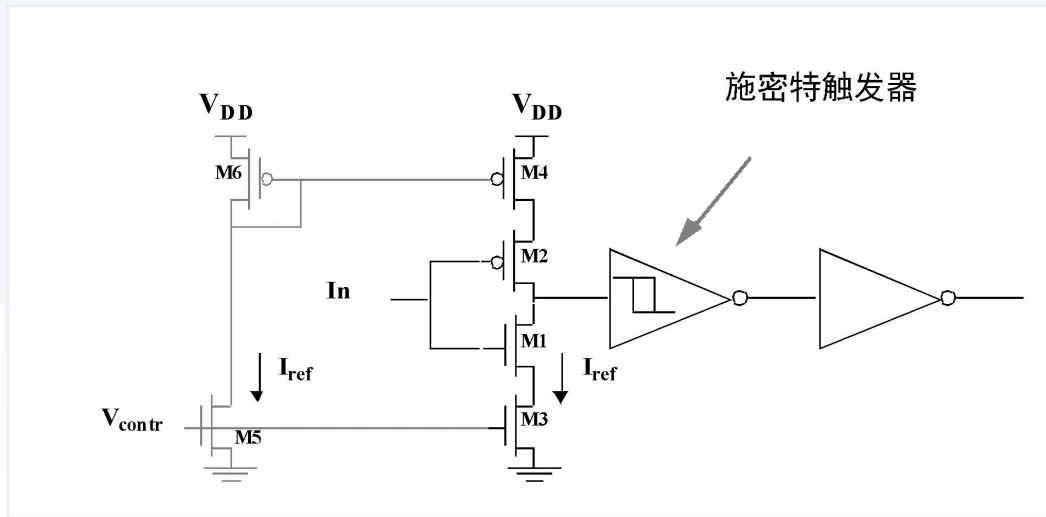
5阶振荡器

弛张式振荡器



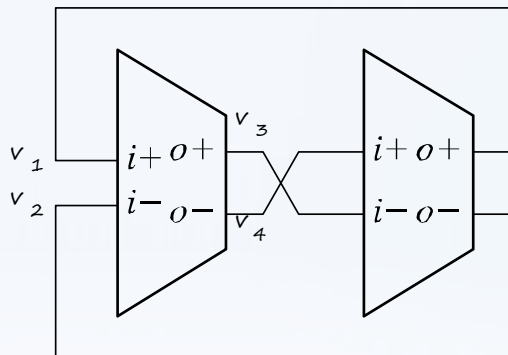
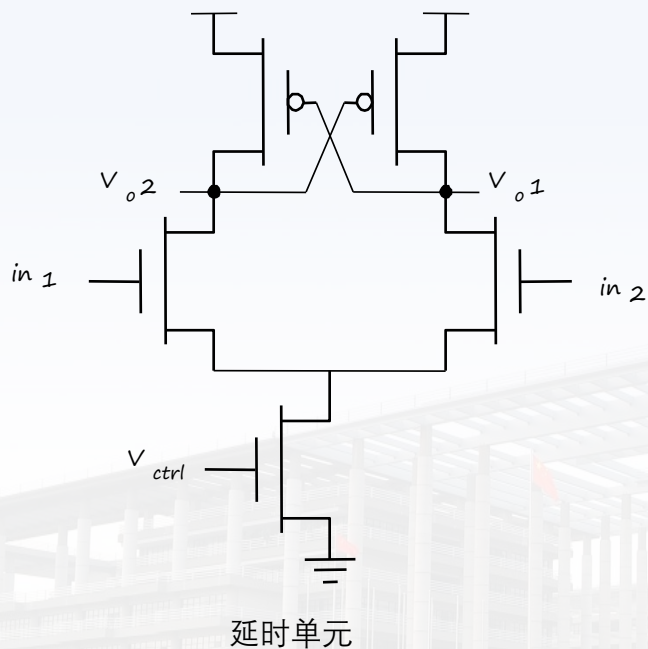
$$T = 2 (\log 3) RC$$

压控振荡器

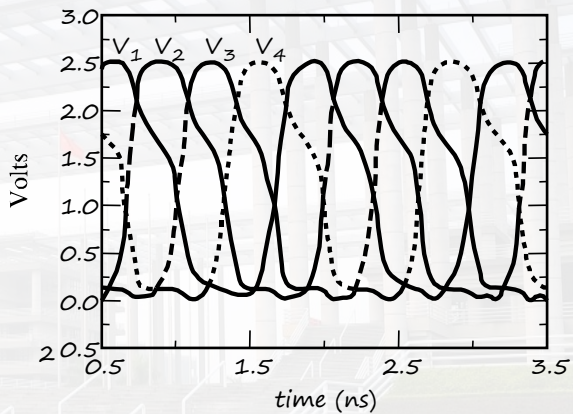


延时与输入电压相关

差分延时单元和压控振荡器



两阶压控振荡器



仿真波形



谢谢！ 请批评指正！



广东工业大学集成电路学院

GUANGDONG UNIVERSITY OF TECHNOLOGY
SCHOOL OF INTEGRATED CIRCUITS